

Deep Delver Balance Pass

Numbers for tile health, pickaxe damage, golem damage, machinery damage, biome pacing, upgrade scaling, and first-pass implementation.

Important pacing note: if each run lasts 60 seconds, then 3 hours equals 180 mining runs maximum, before counting surface/shop time. To make the full game land near 3 hours, the early targets are best treated as cumulative milestones: biome 1 clears around run 10, biome 2 around total run 25, and biome 3 around total run 40.

Main Assumptions

- Each run lasts 60 seconds.
- Starting click cooldown is 0.7 seconds.
- Starting click damage is 1.
- Biome 1 dirt takes 3 clicks.
- A player can realistically mine about 15-22 tiles per first run.
- Later, golems and machinery greatly increase total mining output.
- The player should usually buy the next pickaxe before comfortably entering the next biome.
- Golems and machinery should help, but not instantly replace the pickaxe unless the player specializes into them.

Target Biome Pacing

This pacing keeps the full game around 170-175 active mining minutes, leaving some room for surface/shop time while still feeling like a roughly 3-hour game.

Biome	Layer Name	Target Runs Spent	Total Run Target	Active Mining Time
1	Shallow Dirtworks	10	10	10 min
2	Stone Veins	15	25	25 min
3	Moss Caverns	15	40	40 min
4	Crystal Hollow	16	56	56 min
5	Emberstone Layer	17	73	73 min
6	Ancient Ruins	18	91	91 min
7	Fungal Depths	19	110	110 min
8	Pressure Core	20	130	130 min
9	Astral Geode	22	152	152 min
10	Living Core	18-23	170-175	170-175 min

Material Health Scaling

Each biome uses four material categories: filler, common resource, rare resource, and barrier. With the correct pickaxe tier, filler should usually break in 3-4 clicks, common resources in 4-6 clicks, rare resources in 6-9 clicks, and barriers in 8-12 clicks.

Biome	Filler HP	Common Resource HP	Rare Resource HP	Barrier HP
1	3	5	8	10
2	10	15	22	30
3	24	36	52	70
4	50	75	110	150
5	95	140	210	290
6	170	250	380	520
7	300	450	680	900
8	520	780	1,150	1,550
9	900	1,350	2,000	2,700
10	1,550	2,300	3,500	4,700

Pickaxe Damage

Each pickaxe tier makes the current biome manageable, but the next biome should feel rough until the player upgrades again.

Tier	Pickaxe Name	Biome Unlocked	Click Damage	Same-Biome Filler
1	Rootbound Pickaxe	Biome 1	1	3 clicks
2	Bronze Pickaxe	Biome 2	4	3 clicks
3	Ironbark Pickaxe	Biome 3	9	3 clicks
4	Crystal Pickaxe	Biome 4	18	3 clicks
5	Ember Pickaxe	Biome 5	35	3 clicks
6	Relic Pickaxe	Biome 6	65	3 clicks
7	Spore Pickaxe	Biome 7	115	3 clicks
8	Titanium Pickaxe	Biome 8	200	3 clicks
9	Astral Pickaxe	Biome 9	350	3 clicks
10	Core Pickaxe	Biome 10	600	3 clicks

Pickaxe Upgrade Levels

Each pickaxe can be upgraded several times before the next biome. This lets players push slightly ahead, but the proper tier should still feel like a major power spike.

Upgrade Level	Damage Multiplier
Level 1	1.00x
Level 2	1.20x
Level 3	1.45x
Level 4	1.75x
Level 5	2.10x

Click Speed Progression

Starting click cooldown is 0.7 seconds. Hard-cap normal click speed at 0.30 seconds. A manual specialization can push the cap to about 0.22-0.25 seconds.

Upgrade Level	Click Cooldown
0	0.70s
1	0.64s
2	0.58s
3	0.53s
4	0.49s
5	0.45s
6	0.41s
7	0.38s
8	0.35s
9	0.32s
10	0.30s

Golem Damage

Golems are weaker than pickaxes individually because the player can own several of them. Their strength comes from quantity, upgrades, and specialization.

Tier	Golem Name	Damage Per Hit	Attack Interval	Approx DPS
1	Rootstone Golem	1	1.8s	0.55
2	Bronze Golem	2	1.7s	1.17
3	Ironbark Golem	4	1.6s	2.50
4	Crystal Golem	8	1.55s	5.16
5	Ember Golem	15	1.5s	10.00
6	Relic Golem	28	1.45s	19.31
7	Spore Golem	50	1.4s	35.71
8	Pressure Golem	90	1.35s	66.67
9	Astral Golem	160	1.3s	123.08
10	Core Golem	275	1.25s	220.00

Golem Ownership Limits

Use soft limits to keep early golems from exploding the balance. Stonewarden specialization can increase the maximum active golems.

Game Stage	Suggested Max Active Golems
Biome 1	1
Biome 2	2
Biome 3	3
Biome 4	4
Biome 5	5
Biome 6	6
Biome 7	7

Game Stage	Suggested Max Active Golems
Biome 8	8
Biome 9	9
Biome 10	10

Machinery Damage

Machinery should not just be another golem. It should be slower but stronger, more area-based, and better at scaling through upgrades, fuel, and processing.

Tier	Machine Tier	Damage Per Hit	Attack Interval	Approx DPS
1	Scrap Drill	1	2.8s	0.36
2	Bronze Drill	3	2.7s	1.11
3	Iron Drill	7	2.6s	2.69
4	Crystal Drill	15	2.5s	6.00
5	Ember Drill	30	2.4s	12.50
6	Relic Drill	55	2.3s	23.91
7	Spore Drill	100	2.2s	45.45
8	Titanium Drill	175	2.1s	83.33
9	Astral Drill	300	2.0s	150.00
10	Core Drill	520	1.9s	273.68

Expected Mining Power by Stage

This estimates total mining power when entering each biome. Pickaxe DPS assumes a 0.7 second click cooldown before click speed upgrades.

Biome	Pickaxe DPS Target	Golem DPS Target	Machinery DPS Target	Total Expected DPS
1	1.4	0-0.5	0	1.4-1.9
2	5.7	1-2	0-1	6.7-8.7
3	12.8	4-7	2-4	18-24
4	25.7	10-18	5-10	40-55
5	50	25-40	12-25	85-115
6	93	45-75	25-50	160-220
7	164	90-140	50-100	300-400
8	286	160-250	100-180	550-700
9	500	300-450	180-350	1,000-1,300
10	857	550-800	350-650	1,800-2,300

Biome Completion Requirements

To control run counts, do not rely only on tile HP. Also use a biome progress requirement. A biome can be cleared by reaching a required depth or mining enough depth-progress tiles.

Biome	Target Runs	Target Tiles Per Run	Approx Tiles Needed
1	10	18	180
2	15	28	420
3	15	42	630
4	16	58	930
5	17	78	1,325
6	18	105	1,890
7	19	135	2,565
8	20	170	3,400
9	22	215	4,730
10	18-23	270	4,860-6,210

Resource Distribution

Each biome should be mostly filler. This forces players to dig through ordinary material to reach valuable resources. Use clusters and veins instead of evenly scattering resource blocks.

Biome	Filler	Common Resource	Rare Resource	Barrier
1	72%	22%	4%	2%
2	74%	20%	4%	2%
3	76%	18%	4%	2%
4	78%	16%	4%	2%
5	80%	15%	3%	2%
6	81%	14%	3%	2%
7	82%	13%	3%	2%
8	83%	12%	3%	2%
9	84%	11%	3%	2%
10	85%	10%	3%	2%

Recommended cluster rules: common resources appear in veins of 3-8 blocks, rare resources appear in clusters of 1-4 blocks, and very rare special resources appear as single blocks or guarded pockets.

Upgrade Cost Scaling

Upgrade costs are what really control how many runs each biome takes. Early upgrades should be cheap and frequent. Mid-game upgrades should require planning. Late-game upgrades should require specialized builds.

Pickaxe Tier	Coin Cost	Main Material Cost	Secondary Material Cost
1	50	25	10
2	250	80	30
3	900	180	70
4	2,500	350	125
5	6,500	650	225
6	15,000	1,100	375

Pickaxe Tier	Coin Cost	Main Material Cost	Secondary Material Cost
7	34,000	1,800	600
8	75,000	2,900	950
9	160,000	4,500	1,500
10	350,000	7,000	2,300

Skill Tree and Specialization Multipliers

Keep the general skill tree strong, but not insane. Specialization should push one style much further.

Stat	Max General Bonus
Pickaxe damage	+150%
Click speed	+40%
Golem damage	+150%
Golem speed	+40%
Machine damage	+150%
Machine speed	+40%
Resource gain	+100%
EXP gain	+100%

Specialization	Main Style Bonus
Striker	Manual mining becomes 2.5x-4x stronger
Stonewarden	Golems become 2.5x-4x stronger
Engineer	Machinery becomes 2.5x-4x stronger

Breakpoint Examples

Biome 1

The player has Rootbound Pickaxe, 1 damage, 0.7 second cooldown, no golems at first, and no machines. Biome 1 filler has 3 HP, so filler takes $3 \text{ clicks} \times 0.7\text{s} = 2.1 \text{ seconds}$ per tile. With mistakes and target selection, the player mines around 15-22 tiles per run. Biome 1 requires around 180 tiles of progress, so $180 / 18 = 10$ runs.

Biome 2

The player now has Bronze Pickaxe, 4 damage, maybe 1 Rootstone Golem, and 1-2 click upgrades. Biome 2 filler has 10 HP, so $10 / 4 = 3 \text{ clicks}$. The player mines around 28 tiles per run. Biome 2 requires around 420 tiles of progress, so $420 / 28 = 15$ runs. Total by end of biome 2 is about 25 runs.

Biome 3

The player now has Ironbark Pickaxe, 9 damage, 2-3 golems, maybe Basic Drill, and more click upgrades. Biome 3 filler has 24 HP, so $24 / 9 = 3 \text{ clicks}$. The total system is stronger, so the player mines around 42 tiles per run. Biome 3 requires around 630 tiles of progress, so $630 / 42 = 15$ runs. Total by end of biome 3 is about 40 runs.

Simple Formulas

$$\text{tile_health} = \text{pickaxe_damage_for_that_biome} * \text{desired_clicks}$$

rare_tile_health = pickaxe_damage_for_that_biome * 6 to 9 clicks

barrier_health = pickaxe_damage_for_that_biome * 8 to 12 clicks

Recommended Starting Numbers

Tile HP

Biome 1:	filler 3,	common 5,	rare 8,	barrier 10
Biome 2:	filler 10,	common 15,	rare 22,	barrier 30
Biome 3:	filler 24,	common 36,	rare 52,	barrier 70
Biome 4:	filler 50,	common 75,	rare 110,	barrier 150
Biome 5:	filler 95,	common 140,	rare 210,	barrier 290
Biome 6:	filler 170,	common 250,	rare 380,	barrier 520
Biome 7:	filler 300,	common 450,	rare 680,	barrier 900
Biome 8:	filler 520,	common 780,	rare 1150,	barrier 1550
Biome 9:	filler 900,	common 1350,	rare 2000,	barrier 2700
Biome 10:	filler 1550,	common 2300,	rare 3500,	barrier 4700

Pickaxe Damage

Rootbound Pickaxe: 1
Bronze Pickaxe: 4
Ironbark Pickaxe: 9
Crystal Pickaxe: 18
Ember Pickaxe: 35
Relic Pickaxe: 65
Spore Pickaxe: 115
Titanium Pickaxe: 200
Astral Pickaxe: 350
Core Pickaxe: 600

Golem Damage

Rootstone Golem: 1 every 1.8s
Bronze Golem: 2 every 1.7s
Ironbark Golem: 4 every 1.6s
Crystal Golem: 8 every 1.55s
Ember Golem: 15 every 1.5s
Relic Golem: 28 every 1.45s
Spore Golem: 50 every 1.4s
Pressure Golem: 90 every 1.35s
Astral Golem: 160 every 1.3s
Core Golem: 275 every 1.25s

Drill Damage

Scrap Drill: 1 every 2.8s
Bronze Drill: 3 every 2.7s
Iron Drill: 7 every 2.6s
Crystal Drill: 15 every 2.5s
Ember Drill: 30 every 2.4s
Relic Drill: 55 every 2.3s
Spore Drill: 100 every 2.2s
Titanium Drill: 175 every 2.1s
Astral Drill: 300 every 2.0s
Core Drill: 520 every 1.9s

Important Balance Rule

Do not make the next biome impossible without the new pickaxe. Make it feel bad, not impossible.

Example: a player with a Bronze Pickaxe does 4 damage. Biome 3 filler has 24 HP, so $24 / 4 = 6$ clicks. The player can technically mine biome 3 with a Bronze Pickaxe, but it feels slow. Then they buy the Ironbark Pickaxe and $24 / 9$ becomes about 3 clicks. Now biome 3 feels correct.